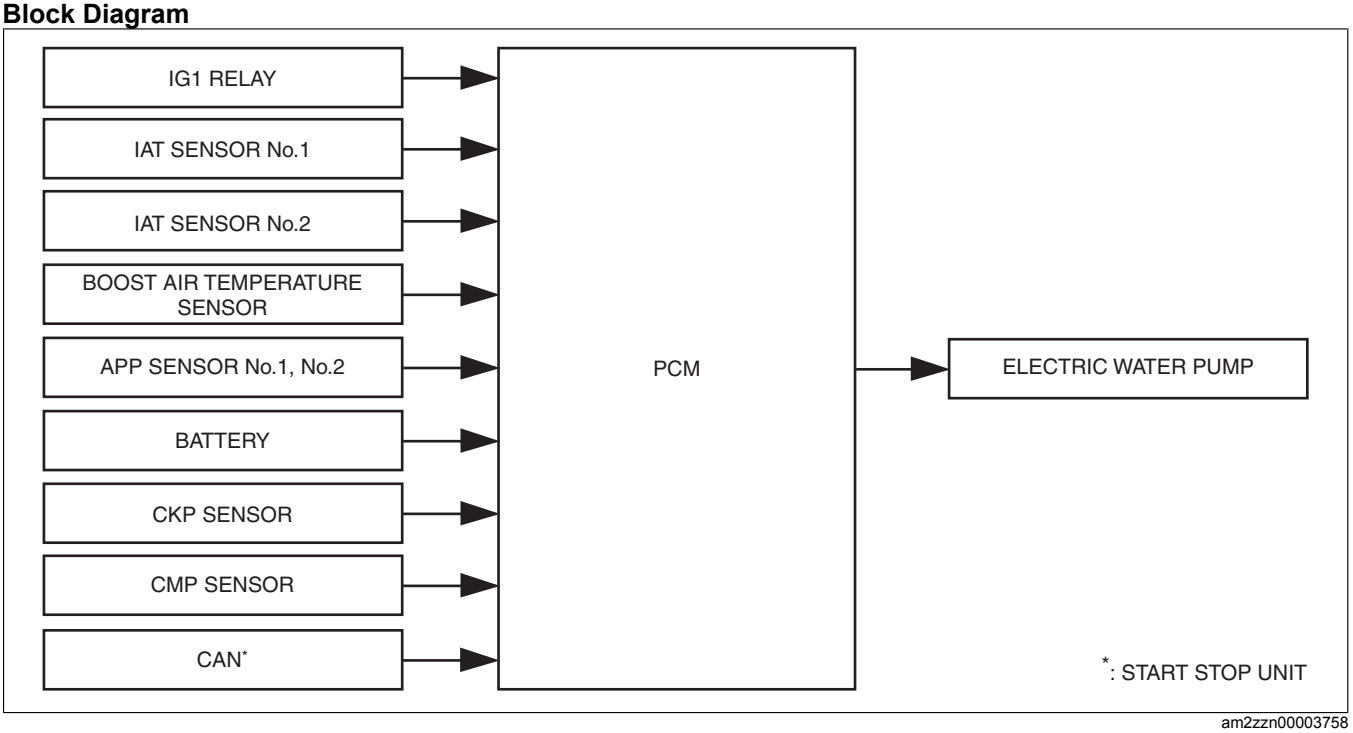
ELECTRIC WATER PUMP CONTROL [SKYACTIV-D 1.5]

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Outline

- The PCM sends a duty signal to the electric water pump according to the engine operation conditions to adjust the water-cooled charge air cooler performance.

Block Diagram



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Operation

- The PCM operates the electric water pump so that the intake air temperature after passing through the water-cooled charge air cooler, detected by the boost air temperature sensor, is the target intake air temperature.

Target intake air temperature

- Changes continuously according to the operation/non-operation of DPF regeneration control and the engine operation conditions (engine speed, fuel injection amount).
- When all of the following conditions are met, the PCM operates the electric water pump.
  - — Battery voltage: 9—16 V
  - — While driving vehicle (condition other than engine start)
  - — Condition is met in which electric water pump operation/non-operation is repeated, or electric water pump operates continuously
- If any one of the above conditions is met, the electric water pump drive is stopped.

Electric water pump operation/non-operation is repeated

- To prevent the water-cooled charge air cooler from over-cooling, the electric water pump operation/non-operation is repeated.
- When the following condition is met, the PCM repeats transmission/non-transmission of the duty signal.
  - — Intake air temperature after passing through water-cooled charge air cooler, detected by boost air temperature sensor, is **5—10 °C {41—51 °F}** higher than target intake air temperature

Electric water pump operates continuously

- To cool the water-cooled charge air cooler rapidly, the electric water pump is operated continuously.
- When the following condition is met, the PCM repeats transmission/non-transmission of the duty signal to the electric water pump.
  - — Intake air temperature after passing through water-cooled charge air cooler, detected by boost air temperature sensor, is **10 °C {50 °F}** higher than target intake air temperature